

Plasmid Classification on DNA sequence with Deep Learning

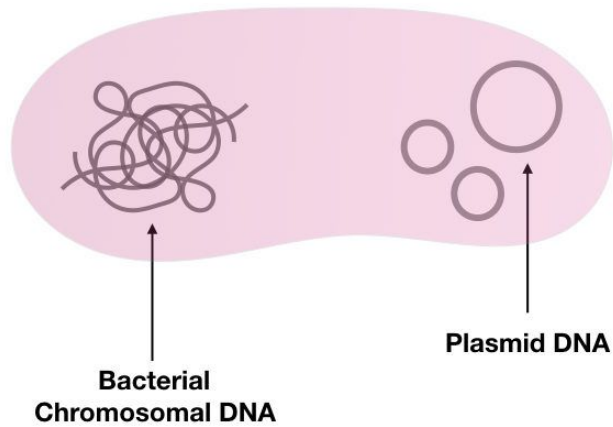
Utilizing Sequences Embedding

Siwakorn Saiphaisri

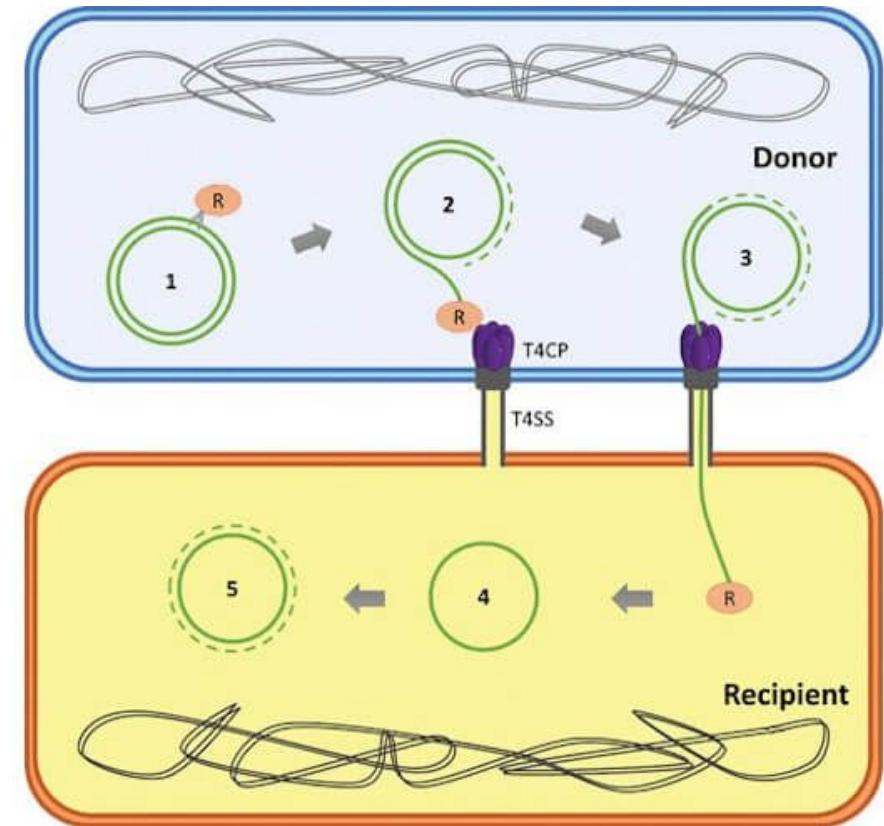
Advisor: Asst. Prof. Dr. Naruemon Pratanwanich



Why Plasmid? AMR and HGT

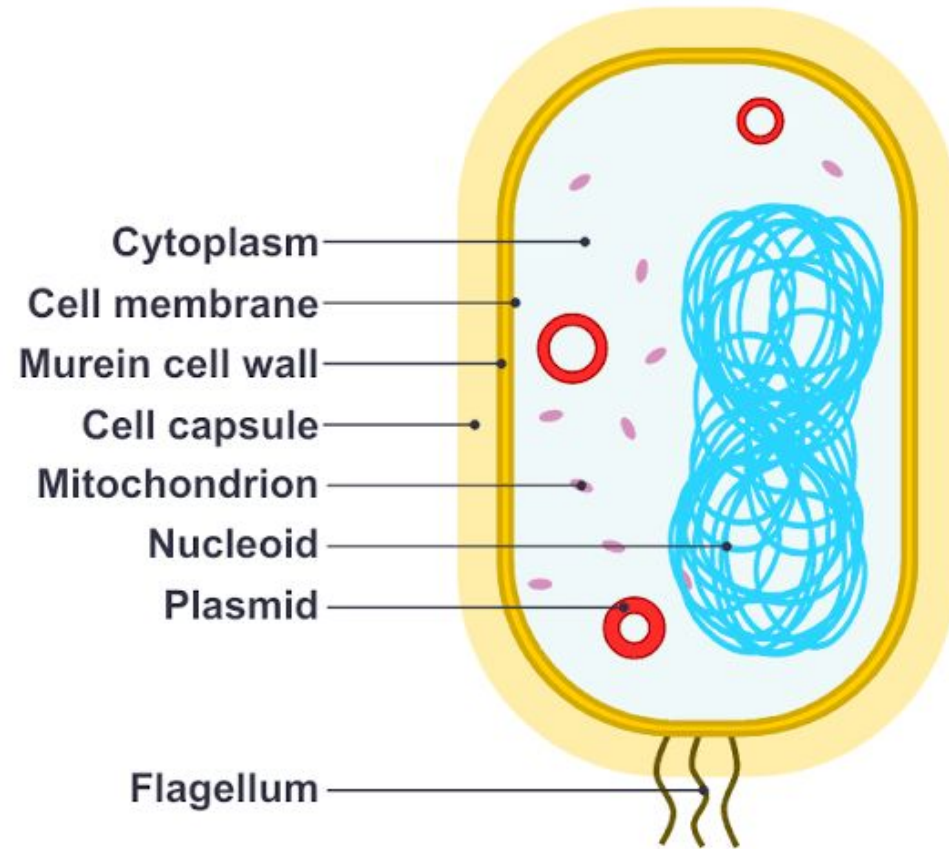


- **AMR Carriers:** Plasmids are key drivers of antimicrobial resistance.
- **HGT Mechanism:** Transfer occurs through Horizontal Gene Transfer between bacteria.
- **Research Goal:** Separating plasmid from chromosome is vital for tracing drug resistance origins.

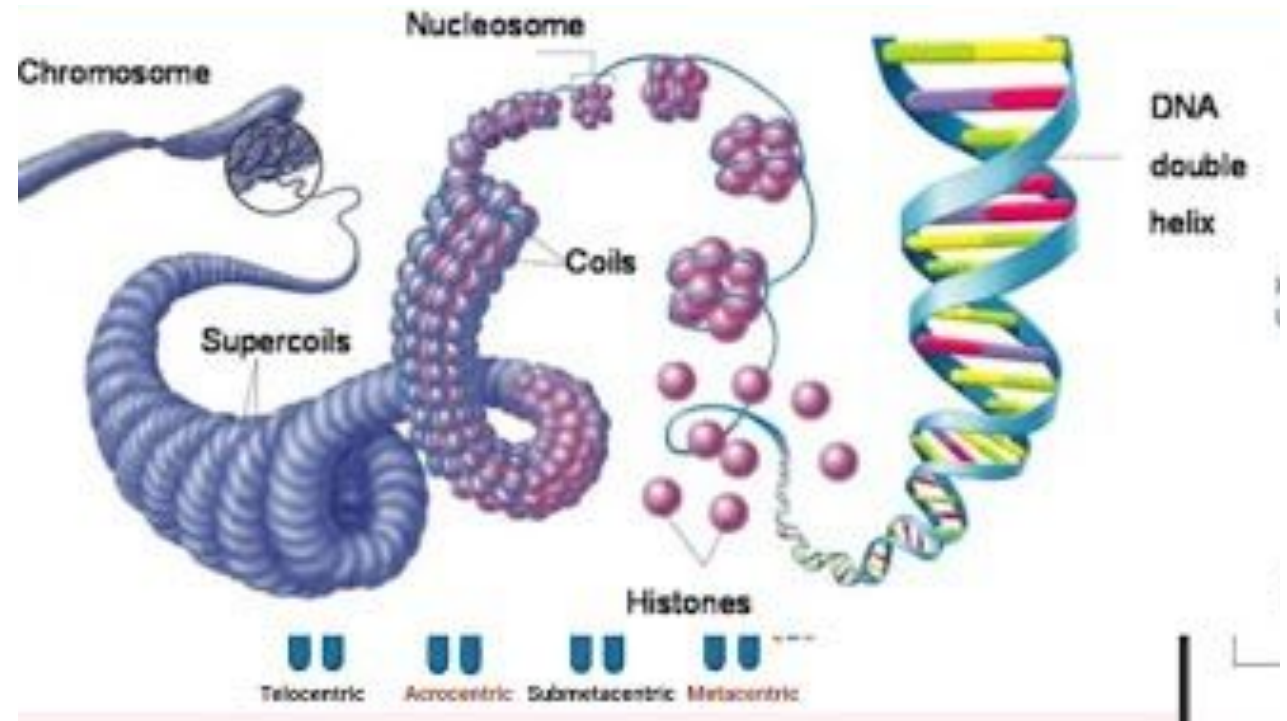


Horizontal Gene Transfer the Antimicrobial Resistant gene

Introduction



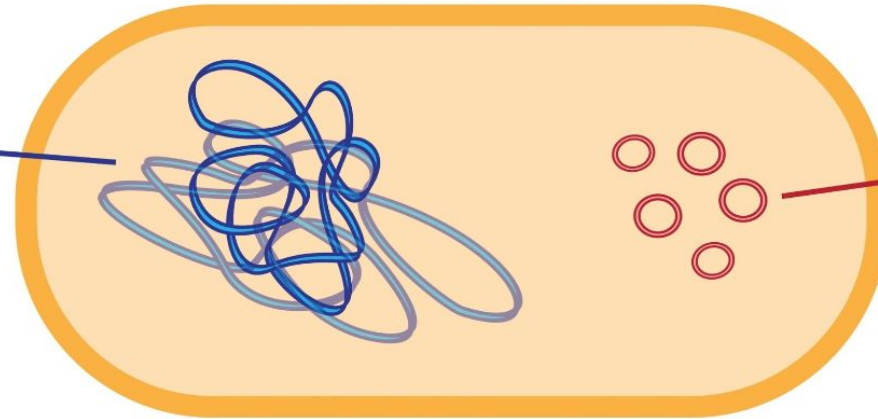
Nucleotide structure (Chromosome/Plasmid)



Introduction

Chromosomal DNA

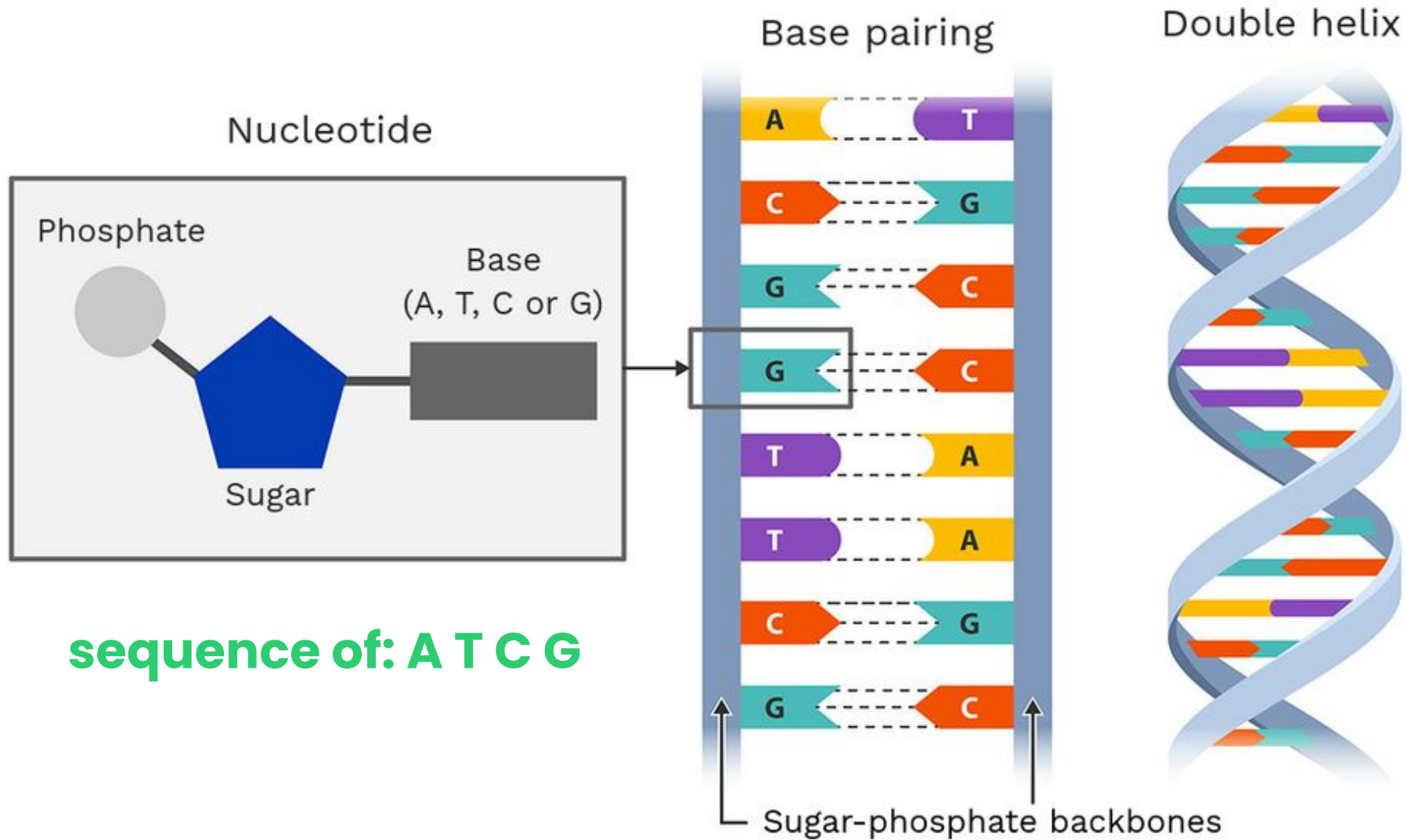
- Genomic DNA that carry all of genetic informations
- Found in both of eukaryote and prokaryote
- Replicate with replication process
- Carry multiple genes for cell survival



Plasmid DNA

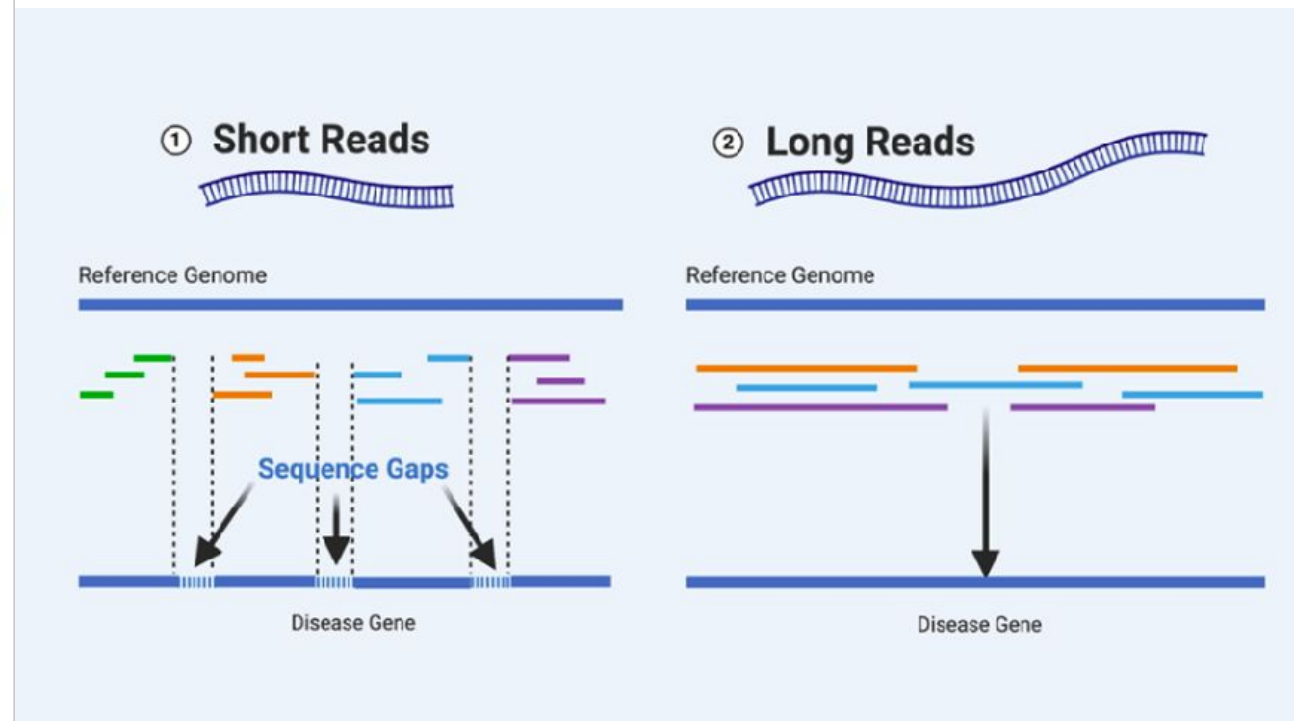
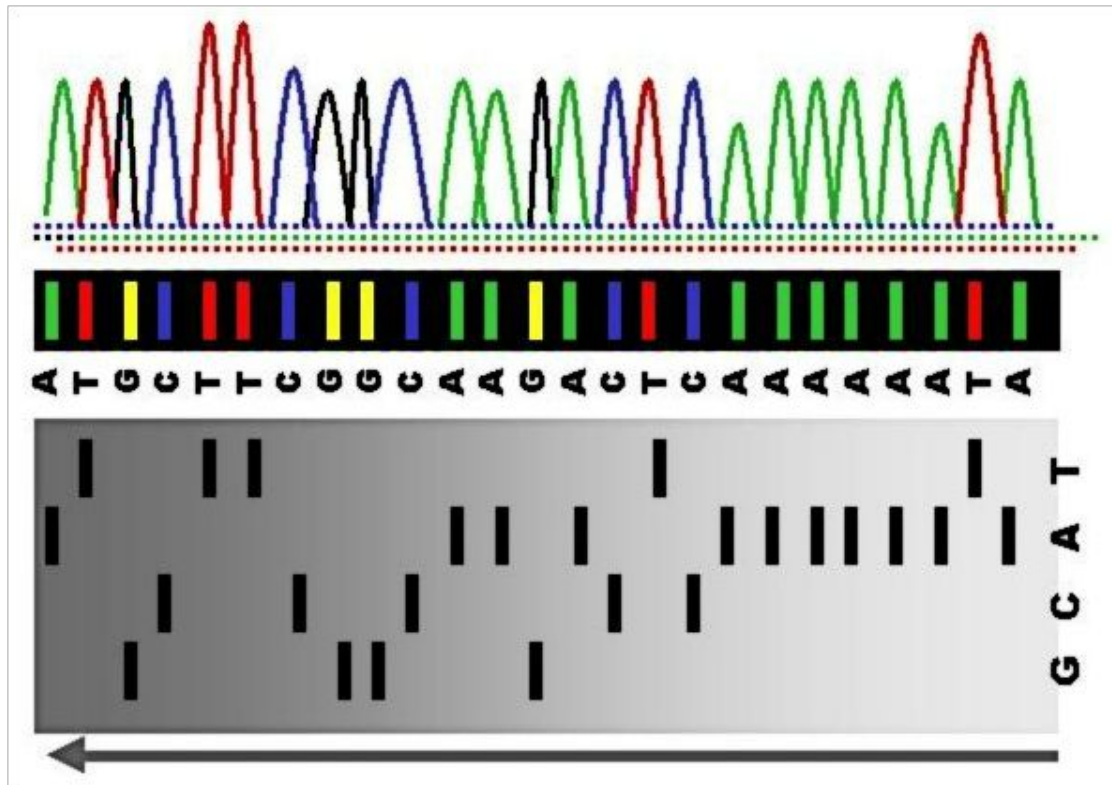
- Extra chromosomal DNA
- Found in prokaryote
- Independent self replication
- Carry genes with special features (antibiotic resistance, drought tolerance, etc.)
- Important for recombinant DNA technology

Introduction



Introduction

DNA Sequencing



Introduction

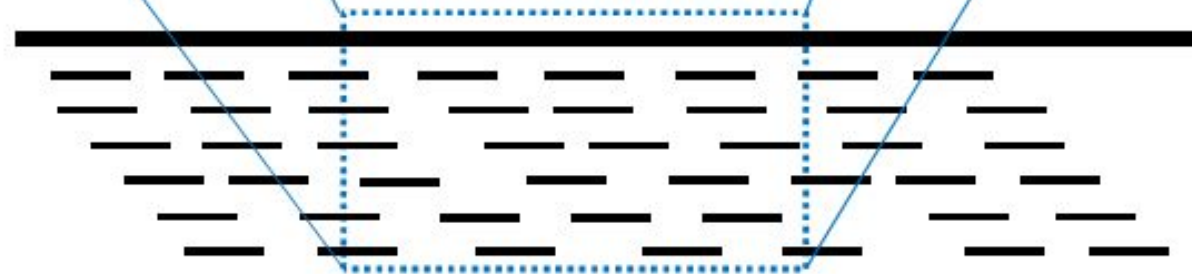
Alignment



AGGGTTATTTACCCTACTGCGACTATCTAGT				
AGG	ATTAC	ACTGCG	TAGCTA	T
AGGG	TTACCC	CTGCGA	AGCTAG	
AGGGT	TCCTAC	GCGACT	GCTAGT	
GGGTTA	CCTAC	ACTAGC	GT	
AGG	ATTAC	TACTGC	AGCTAG	
AGGGT	TTACCC	CGACTAG	AGT	

Reference

Short
Reads



Introduction

Reads



Contigs



```
>AT1G09780 | 1 | training
GTGGAGTAGAAGAATTGAGAGCCTTATCAG
TTTTTGAAGAGAGGGCTGAAACTCTCTAGT
TATCTTTTGTGCTTTTCTAATAATAAGAG
TTTACACACAG
>AT1G31812 | 0 | testing
TCCTCATCTGCAGTAACTTTATCTTAAGCA
TCAAATAACATTGCATAAGACTTGTTCTT
GCTCTTGTTTCTATCATATTTAAGCTAT
CTACTTTGTGA
```

Part 1

Part 2

Part 3

Contigs

Connected sequence of reads

Complete Sequence

The contigs that completely to represent whole sequence

Fasta

The file of sequences in a single file can be found many complete sequences

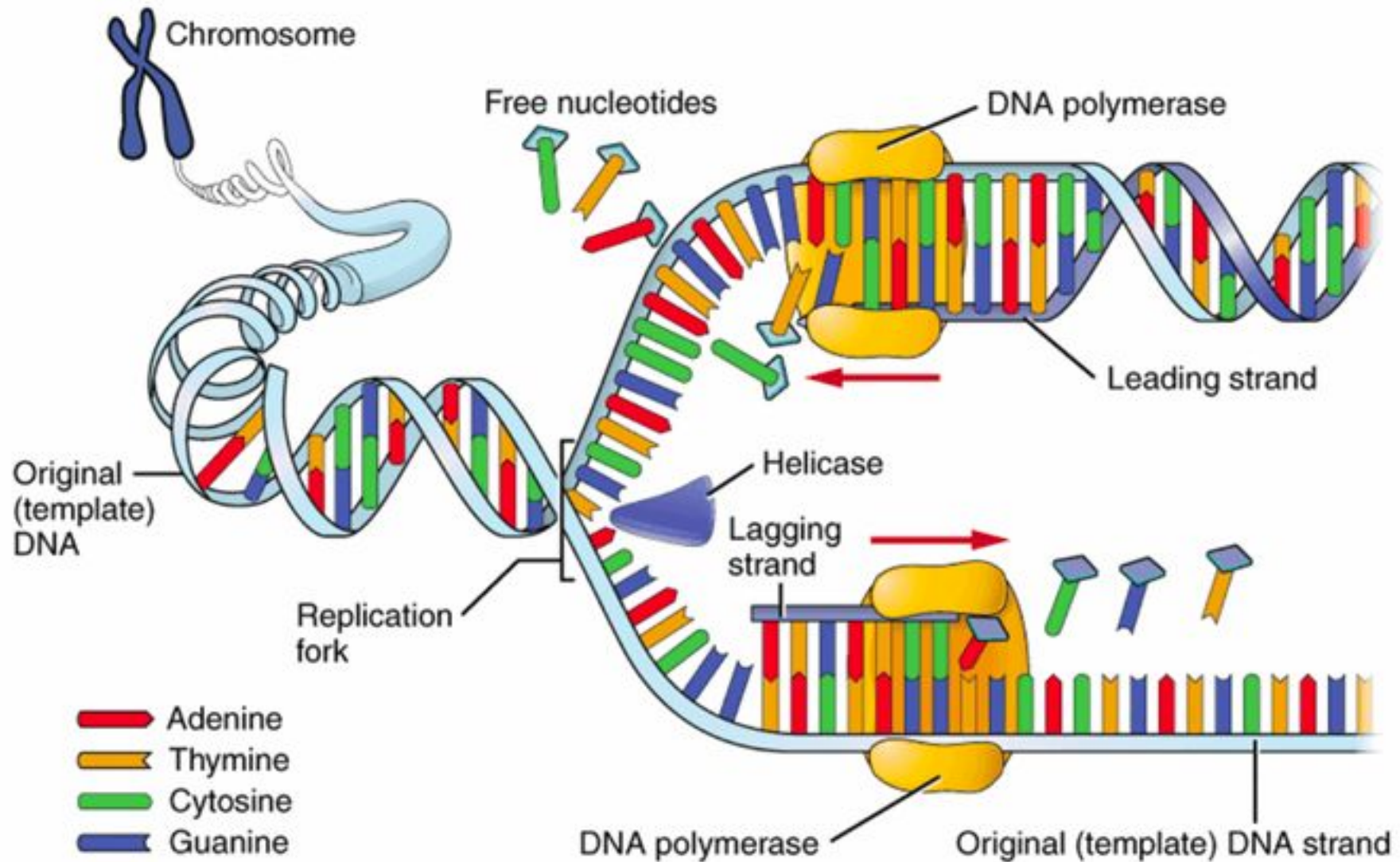
Introduction (Fasta)

Format of a FASTA definition line

>Seq1 [organism=Carpodacus mexicanus] [clone=6b] actin (act) mRNA, partial CDS
CCTTTATCTAATCTTTGGAGCAYGAGCTGGCATAGTTGGAACCGCCCTCAGCCTCCTCATC

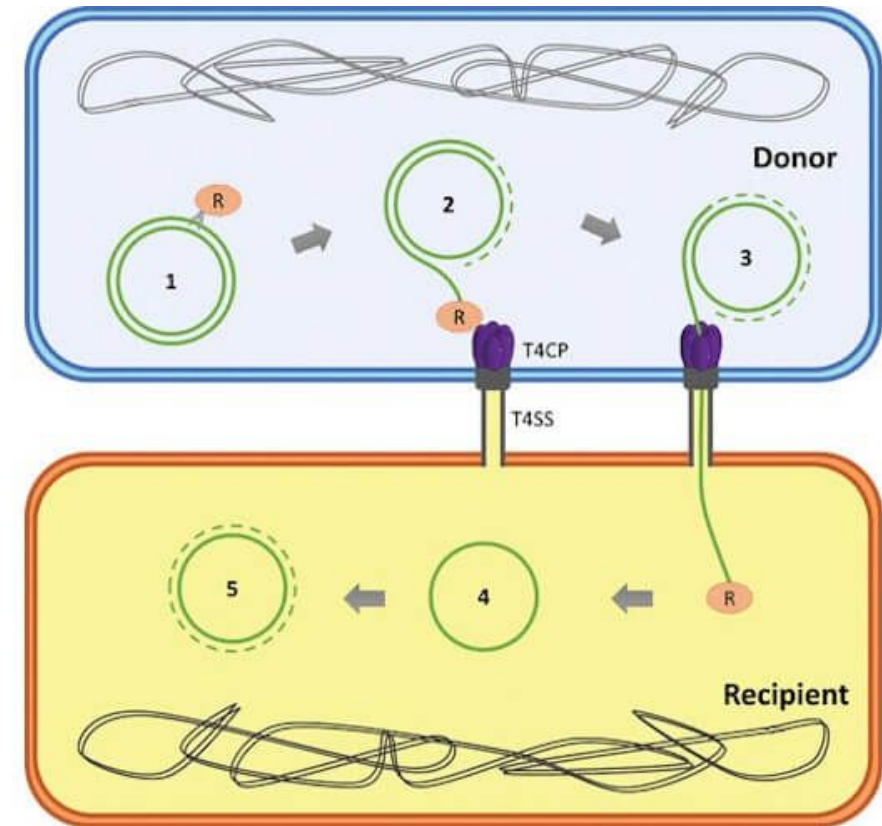
[illegible]

Introduction



Introduction

- **Antimicrobial Resistant Gene Carriers:**
Plasmids are key drivers of antimicrobial resistance.
- **Horizontal Gene Transfer**
Mechanism: Transfer occurs through Horizontal Gene Transfer between bacteria.



Horizontal Gene Transfer the Antimicrobial Resistant gene

Project Objectives

Classify between Chromosome & Plasmid

Develop the classifier model of Plasmid and Chromosome in Bacteria genome by using **Sequence Embedding** technique.

Challenge in this task

High computational power required

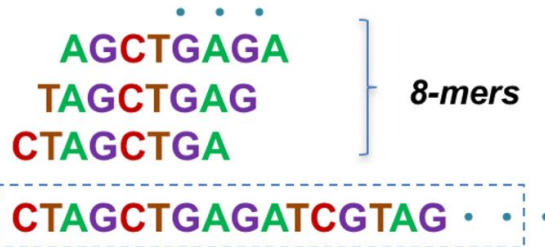
The sequence length is not consistent from our dataset they are in range of **1kbps to 100kbps**. And many bioinformatics tools are **required huge storage and memory** as well.

Exist Methods & Limitation in ML technique



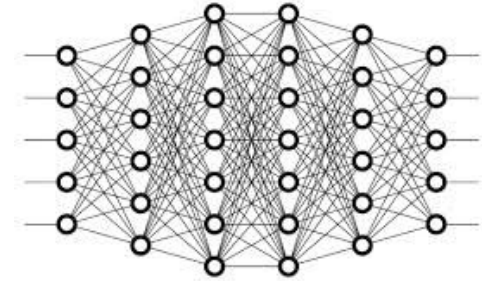
Local Database

Demands large storage, high computation, and significant time overhead.



k-mer Frequency

Suffers from context loss and ignores biological structure or order.

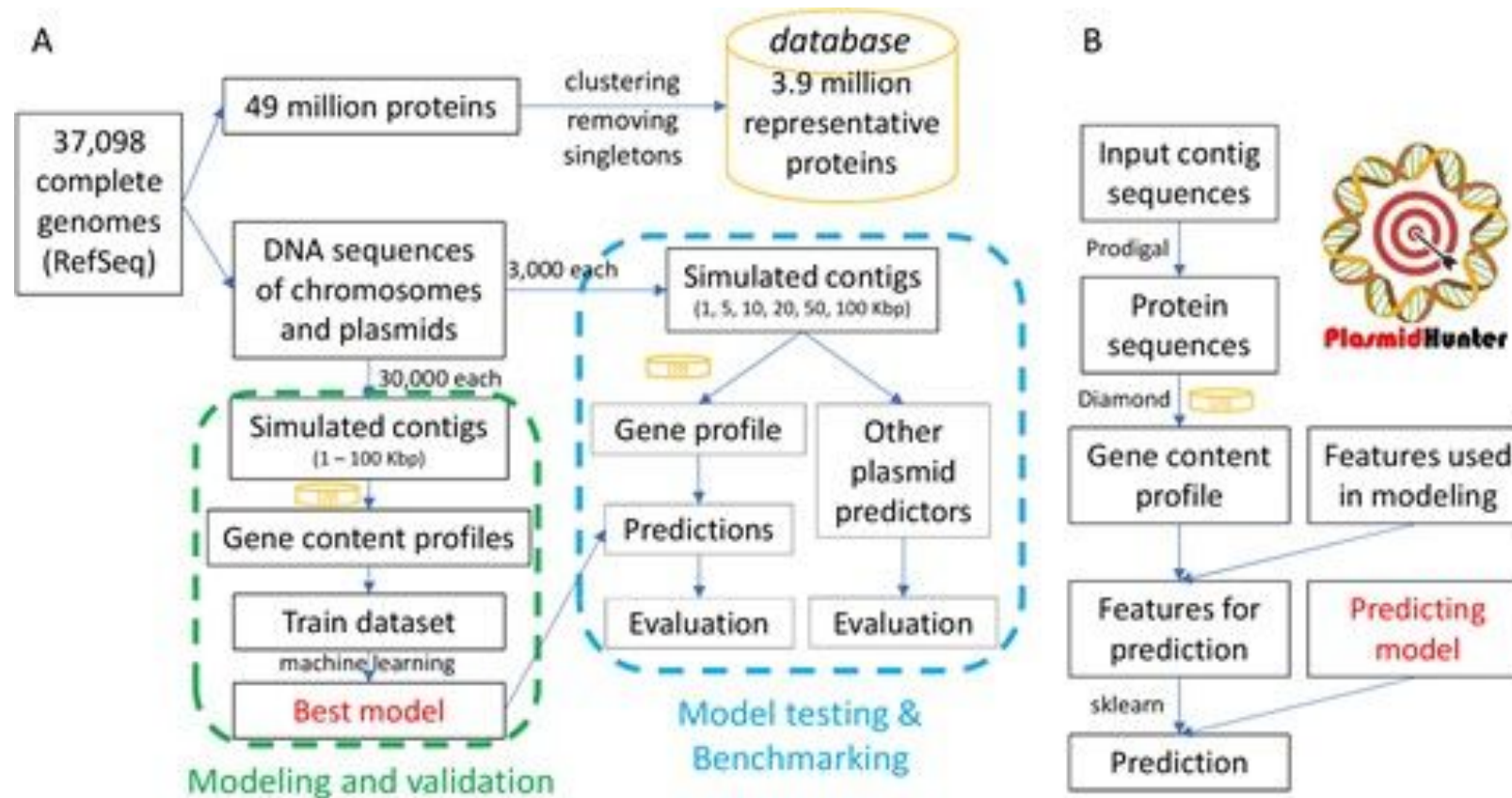


Deep Learning

More Challenge to train with huge data

Exist Methods & Limitation in ML technique

Protein Clustering to Naive Bayes classifier



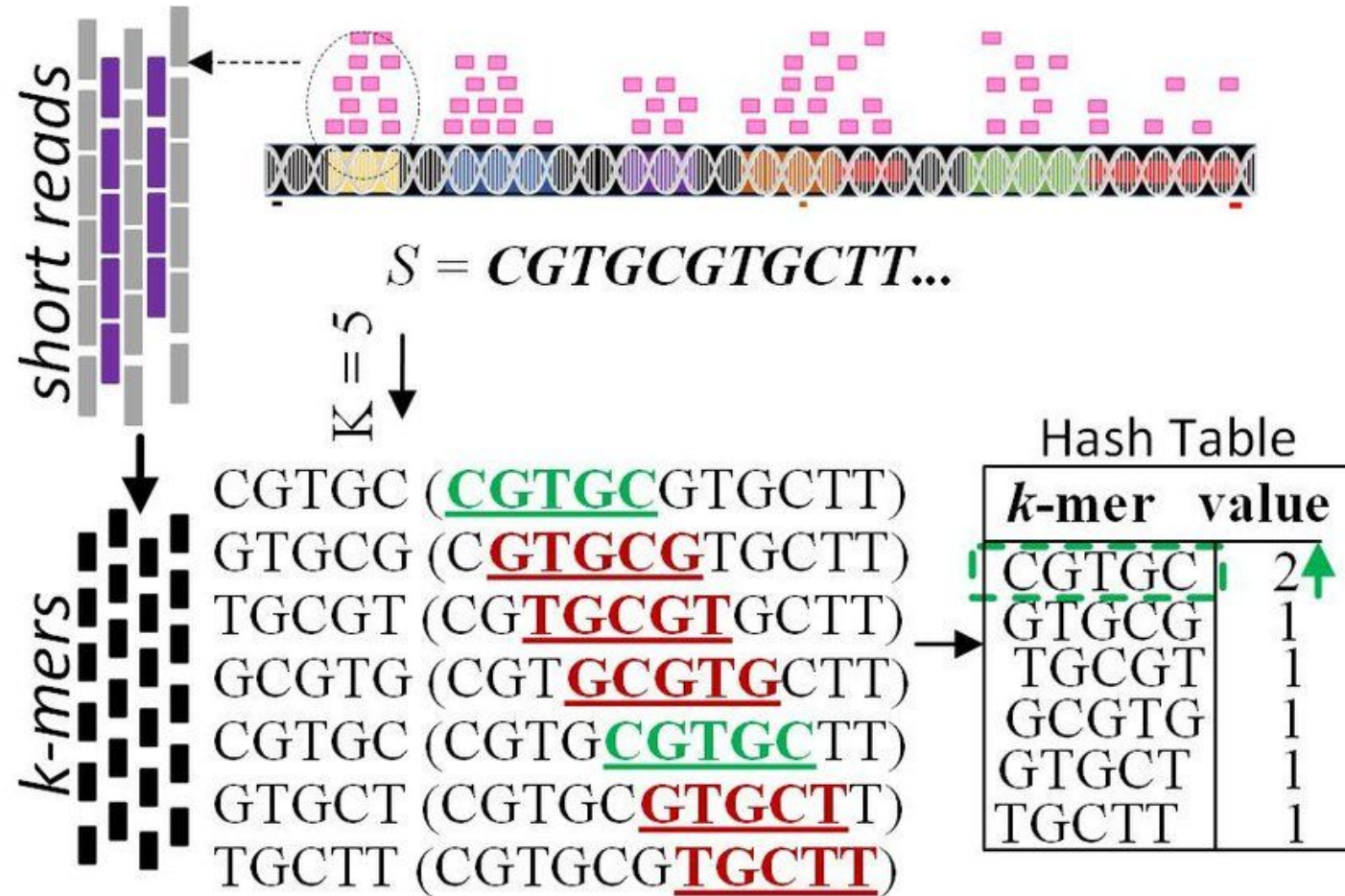
Advantage

Fast, Easy to train and low computational power required

Disadvantage

Context loss. Ignore Biological structure and order.

Exist Methods & Limitation in ML technique



mlplasmids



Advantage

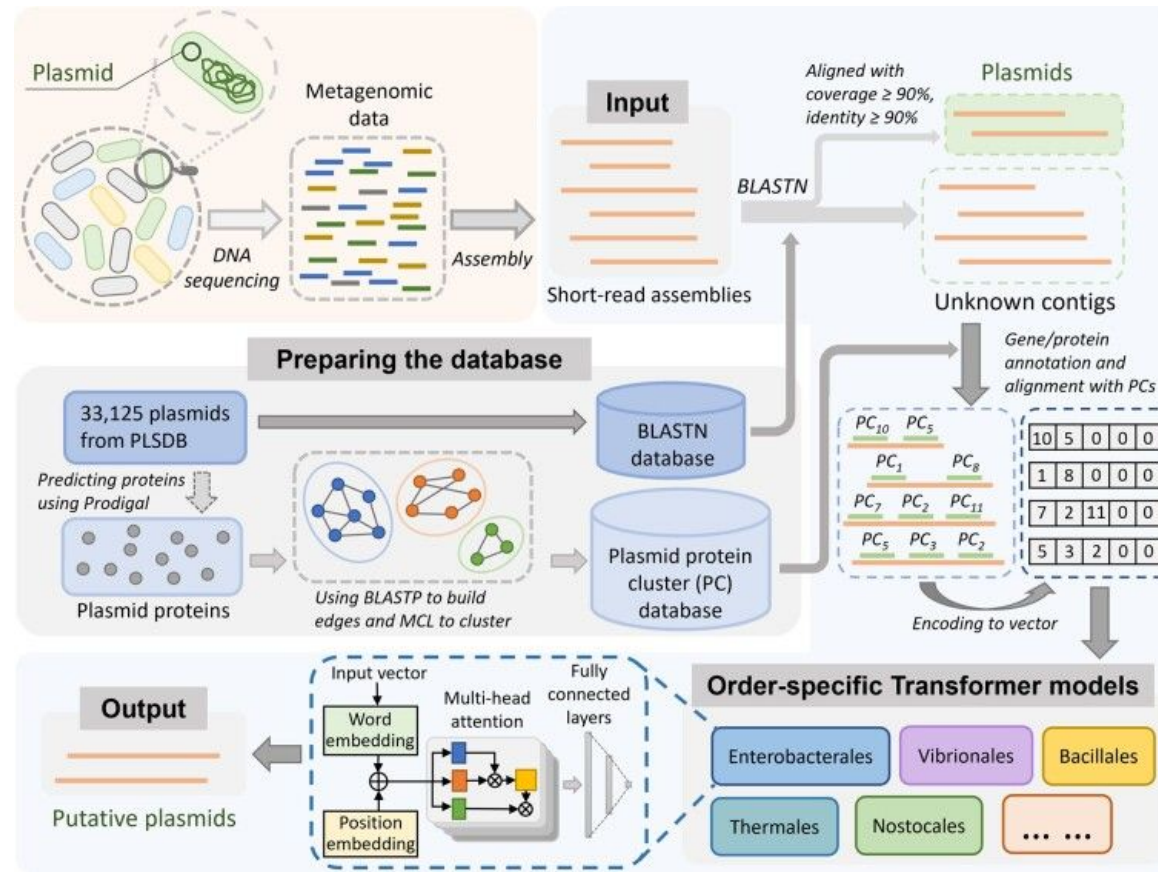
Fast, Easy to train and low computational power required

Disadvantage

Context loss. Ignore Biological structure and order.

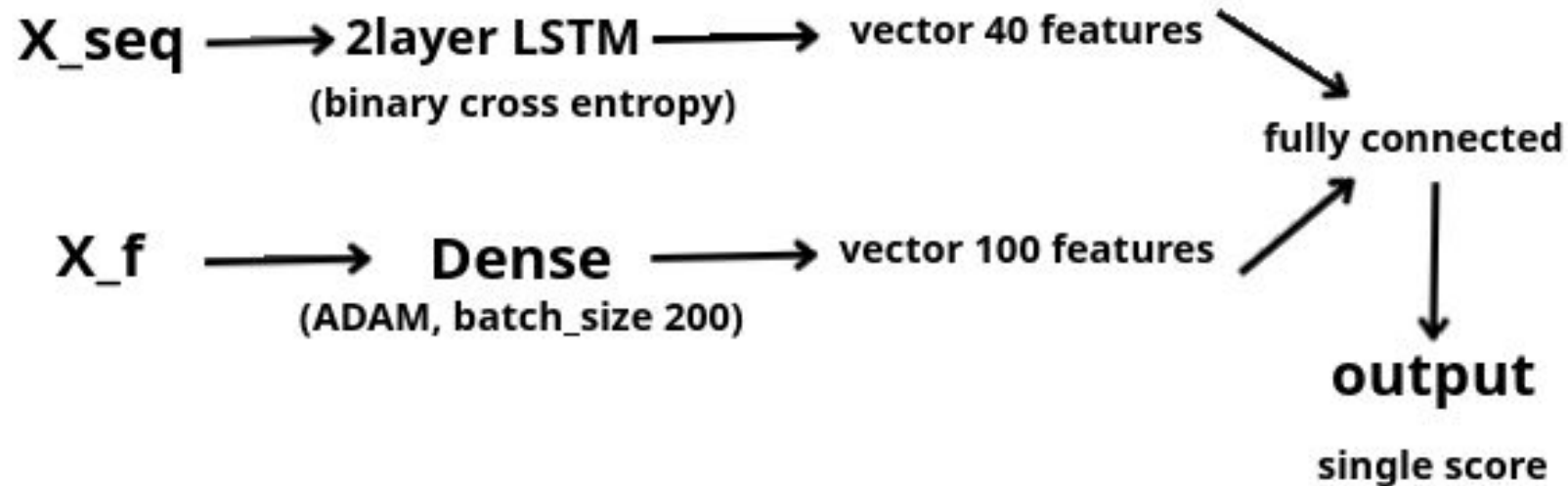
Exist Methods & Limitation in ML technique

Hybrid Structure Blast alignment DB with Transformer Model



Exist Methods & Limitation in ML technique

Ensemble model with encoded sequence and tabula feature



Advantage

Train with many data format

Disadvantage

Complexity and Time consume

Expectation of our propose

Protein/Function Pipeline

DNA sequence → Extract Protein → Embedding → Trainable classifier

Advanced Problem Solving

Mitigate issues related to context loss and structural noise inherent in previous DNA classification models.

Lightweight Architecture

Reduce dependency on massive databases and heavy computational resources for faster classification.

Why This Functional Approach?

“Protein sequence is biological language”

DNA

AATGCGTAA....CTATA



Protein

HGTDDDBFGSTDHJKI



Classify

Plasmid / Chromosome



Alike Sentiment Analysis Model

Dataset Source

National Database of Biological Data

NCBI refseq

The Reference Sequence (RefSeq) collection provides a comprehensive, integrated, non-redundant, well-annotated set of sequences

Plasmid collected only from NCBI

PLSDB

plasmid records that are collected from the NCBI database and subsequently processed to remove incomplete, inconsistent or chromosomal

Dataset Source

NCBI refseq

The Reference Sequence (RefSeq) collection provides a comprehensive, integrated, non-redundant, well-annotated set of sequences

Index of /refseq/release/bacteria

Name	Last modified	Size
Parent Directory		-
bacteria.1.1.genomic.fna.gz	2026-03-20 04:29	293M
bacteria.1.2.genomic.fna.gz	2026-03-20 04:29	64M
bacteria.1.genomic.gbff.gz	2026-03-20 04:29	337M
bacteria.10.1.genomic.fna.gz	2026-03-20 04:29	292M
bacteria.10.2.genomic.fna.gz	2026-03-20 04:29	58M

Index of /refseq/release

Name	Last modified	Size
Parent Directory		-
announcements/	2026-03-19 15:47	-
archaea/	2026-03-19 16:12	-
bacteria/	2026-03-20 08:01	-
complete/	2026-03-20 03:26	-
fungi/	2026-03-19 18:20	-
invertebrate/	2026-03-19 16:50	-
mitochondrion/	2026-03-19 15:47	-
other/	2026-03-19 18:22	-
plant/	2026-03-19 16:10	-
plasmid/	2026-03-19 18:22	-
plastid/	2026-03-19 18:20	-
protozoa/	2026-03-19 18:23	-
release-catalog/	2026-03-20 08:02	-
release-error-notice/	2026-03-19 15:46	-
release-notes/	2026-03-20 11:50	-
release-statistics/	2026-03-20 08:01	-
vertebrate_mammalian/	2026-03-20 04:29	-
vertebrate_other/	2026-03-19 18:15	-
viral/	2026-03-19 15:47	-
README	2026-03-19 15:46	18K
RELEASE_NUMBER	2026-03-20 08:02	4

Dataset Source

PLSDB

plasmid records that are collected from the NCBI database and subsequently processed to remove incomplete, inconsistent or chromosomal

The plasmid database PLSDB

This resource is database of plasmids by NCBI.
Plasmids can be explored and filtered using the [browsing](#) tables.
Nucleotide sequences can be [searched](#) in the database by either using [Mash](#) for longer sequences (e.g. contigs or long reads), or by using [BLASTn](#) for shorter sequences, e.g. genes.

72360
Records

6027698
Annotations

250691
Predic. Virulence Factors

12796
Predicted BGCs

Dataset Setup

All Exist

Train Data (from NCBI refseq)

- Potential Plasmids: 15,604 files
- Potential Chromosomes: 19,836 files

Test Data (from PLSDB)

- Plasmidomes: 3,416 seqs
- Simulated Contigs: 24,744 seqs

Current Usage (with low computational resource)

Development Data (from NCBI refseq)

- Potential Plasmids: 250 files
- Potential Chromosomes: 250 files

Test Data (from PLSDB)

- Plasmidomes: 500 seqs
- Simulated Contigs: 500 seqs

System Architecture Overview

Protein/Function Pipeline

DNA → ORFs → ESM-2 Embedding → Attention

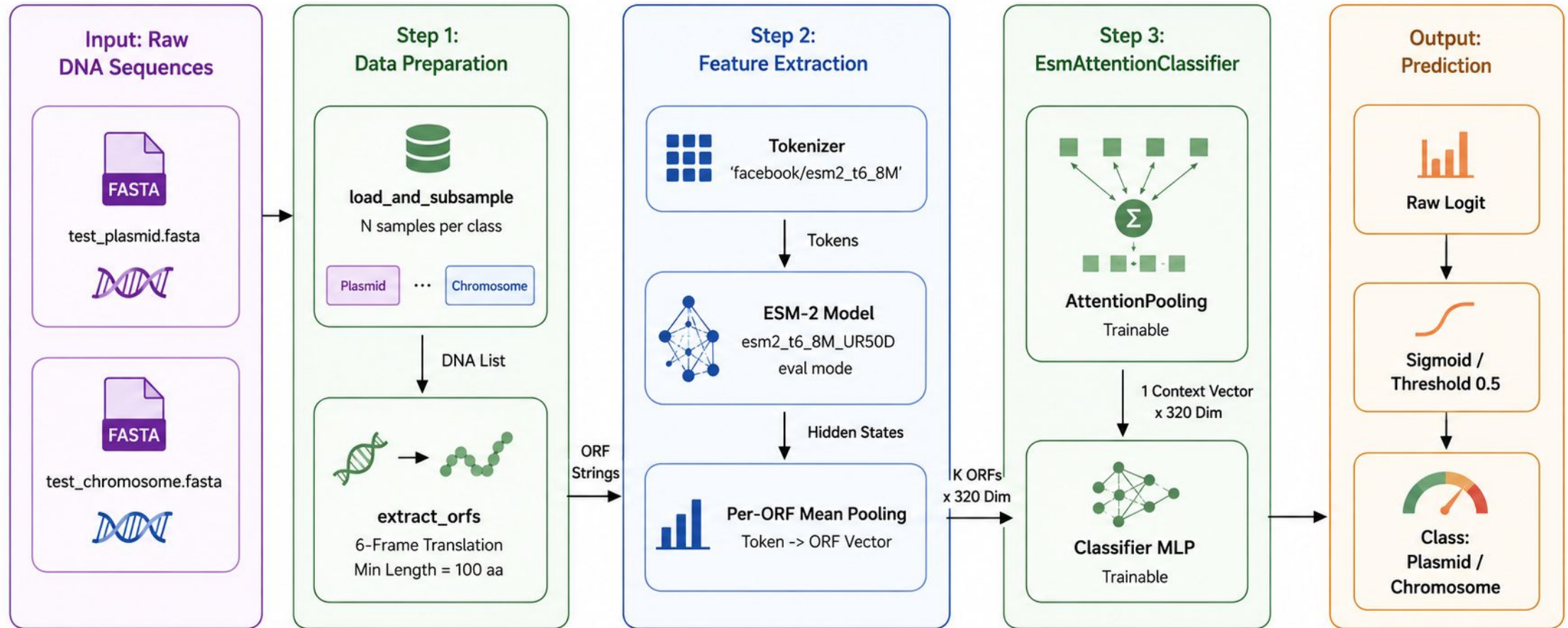
DNA/Structure Pipeline

DNA → DNA-BERT → Attention

Fusion: Concat & MLP for Final Classification

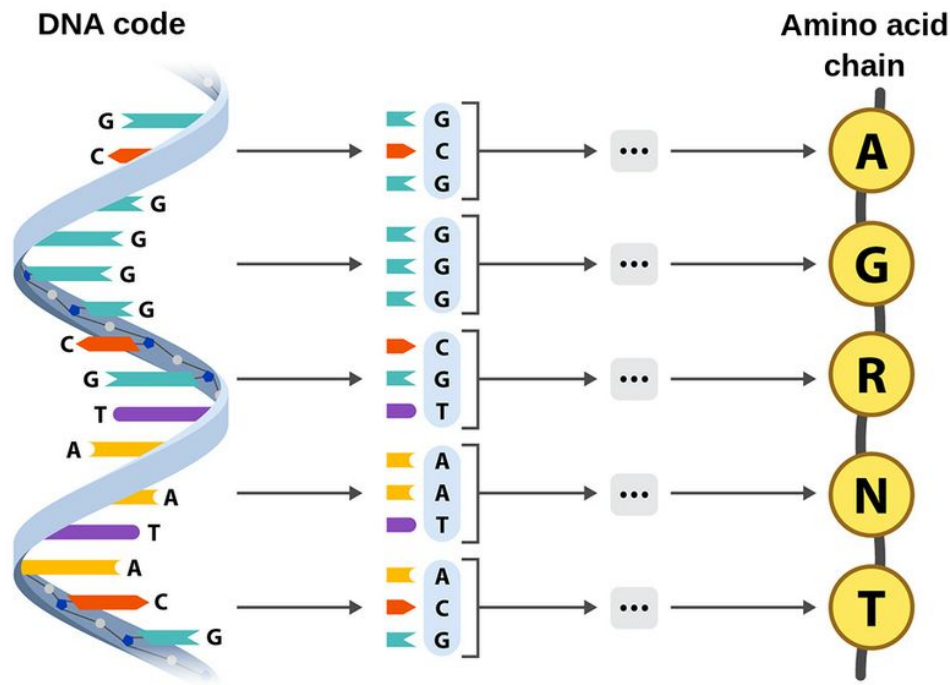


System Architecture Overview



Protein Open Reading Frame (ORFs)

- **Input:** DNA sequence of different 4 characters (T,C,A,G)
- **Output:** Protein Sequence of different 20 characters (Amino Acid)



		second base in codon							
		T	C	A	G				
T	TTT	Phe	TCT	Ser	TAT	Tyr	TGT	Cys	T
	TTC	Phe	TCC	Ser	TAC	Tyr	TGC	Cys	C
	TTA	Leu	TCA	Ser	TAA	stop	TGA	stop	A
	TTG	Leu	TCG	Ser	TAG	stop	TGG	Trp	G
C	CTT	Leu	CCT	Pro	CAT	His	CGT	Arg	T
	CTC	Leu	CCC	Pro	CAC	His	CGC	Arg	C
	CTA	Leu	CCA	Pro	CAA	Gln	CGA	Arg	A
	CTG	Leu	CCG	Pro	CAG	Gln	CGG	Arg	G
A	ATT	Ile	ACT	Thr	AAT	Asn	AGT	Ser	T
	ATC	Ile	ACC	Thr	AAC	Asn	AGC	Ser	C
	ATA	Ile	ACA	Thr	AAA	Lys	AGA	Arg	A
	ATG	Met	ACG	Thr	AAG	Lys	AGG	Arg	G
G	GTT	Val	GCT	Ala	GAT	Asp	GGT	Gly	T
	GTC	Val	GCC	Ala	GAC	Asp	GGC	Gly	C
	GTA	Val	GCA	Ala	GAA	Glu	GGA	Gly	A
	GTG	Val	GCG	Ala	GAG	Glu	GGG	Gly	G
		third base in codon							

Embedding and Attention

Model Variant	Layers	Embedding Dim (dmodel)	Attention Heads	Parameters
esm2_t6_8M	6	320	20	8 Million
esm2_t12_35M	12	480	20	35 Million
esm2_t30_150M	30	640	20	150 Million
esm2_t33_650M	33	1280	20	650 Million
esm2_t36_3B	36	2560	40	3 Billion
esm2_t48_15B	48	5120	40	15 Billion

ESM-2 Model

1. ESM-2 Pretrained

Understands biological function by encoding proteins into vector embeddings.

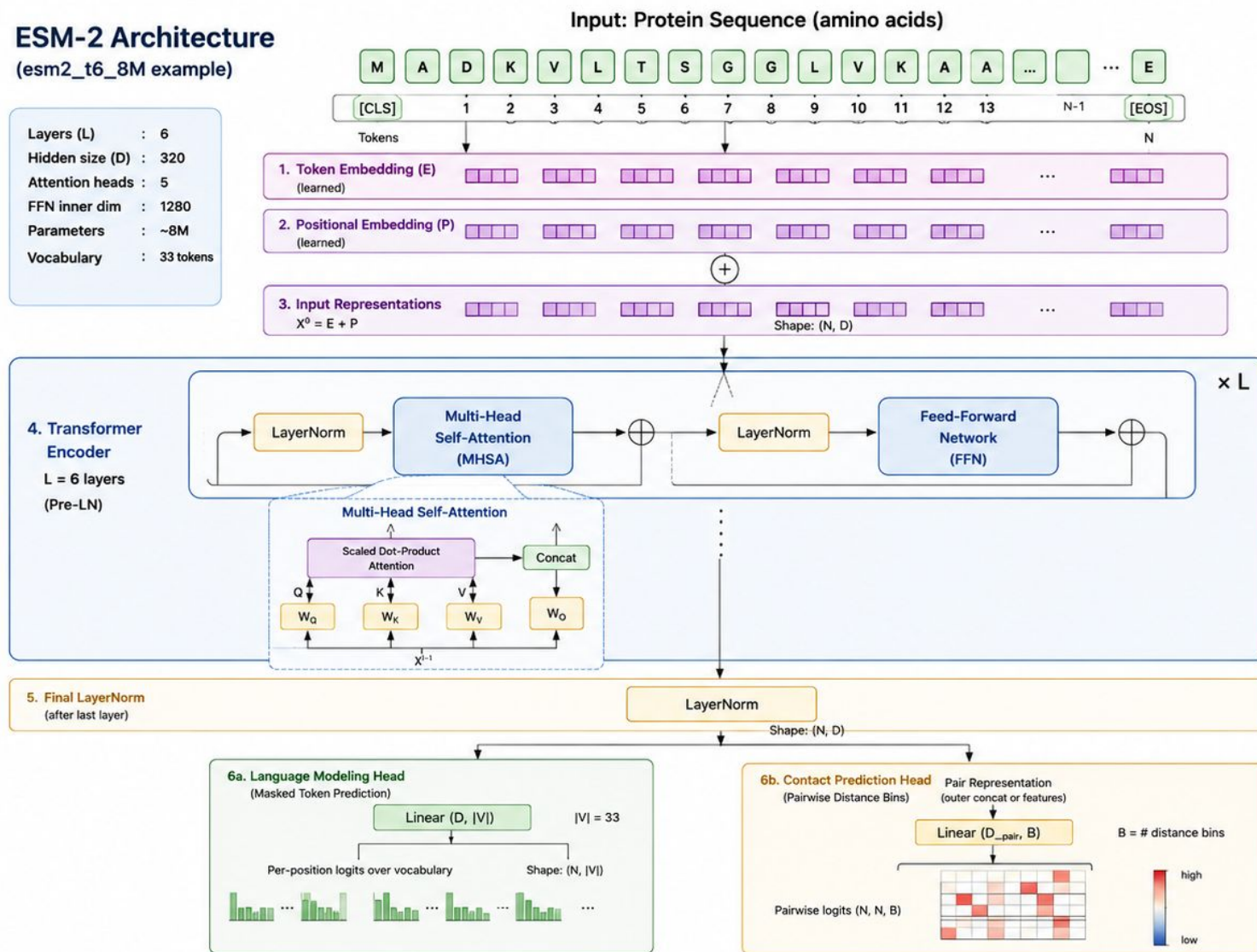
2. Attention Pooling

Identifies "Signature Genes" by focusing on high-weight components while ignoring noise.

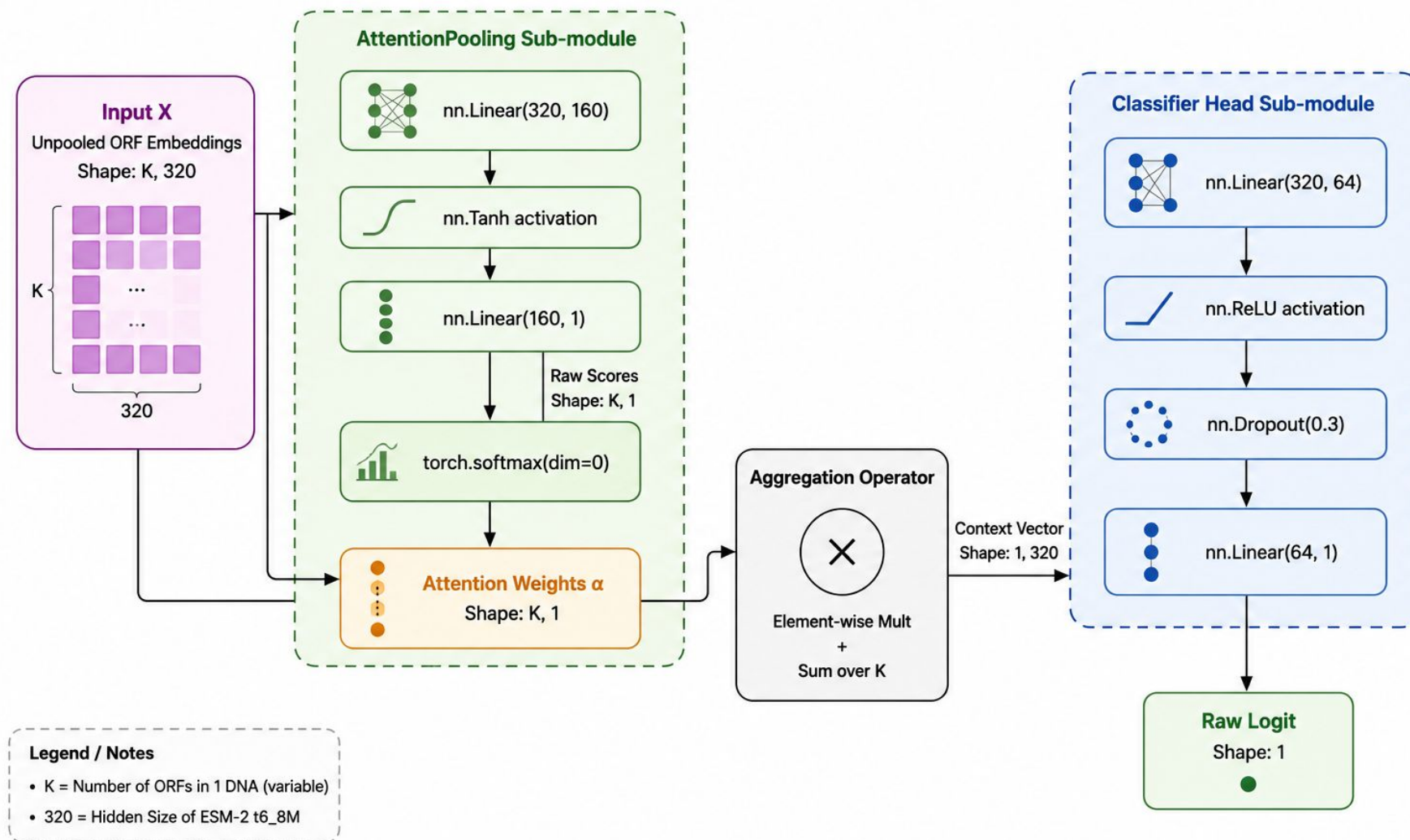
Embedding and Attention

ESM-2 Architecture (esm2_t6_8M example)

Layers (L) : 6
Hidden size (D) : 320
Attention heads : 5
FFN inner dim : 1280
Parameters : ~8M
Vocabulary : 33 tokens



Embedding and Attention



Preliminary Result of Baseline Model (Latest Progress)

- **Baseline :** We selected baseline model of various technique to compare with our propose.
- **Evaluation:** Benchmark data of simulate contigs and plasmid with various length in range 1kbp to 100kbp.

Method	Core Concept	Accuracy	Precision	recall	F1-score
PlasClass	k-mer based	0.8580	0.8537	0.8640	0.8588
mlPlasmid	SVM + k-mer	0.5510	0.5515	0.5510	0.5498
deeplasmid	Deep Learning	0.8681	0.9917	0.7422	0.8490

Preliminary Result of Baseline Model

- **Training :** From limited computational power we using 50 fasta files of Plasmid and 50 fasta files for Chromosome.
- **Evaluation:** Benchmark data of simulate contigs and plasmid with various length in range 1kbp to 100kbp.

Method	Core Concept	Accuracy	Precision	recall	F1-score
Model A	ESM-2 based	0.7431	-	-	0.7619

Next Step

- **Increase computational power scale:** Access server devices
- **Experiment:** Experiment various combination as a Ensemble architecture
- **Develop Model:** Explore more experiment setup by adjust trainable architecture and customize hyperparameters tuning

Q&A

Thank you for your
attention!